Yixuan Wang

Ph.D. Student, Mechanical & Aerospace Engineering · University of Florida · Expected May 2027

Gainesville, FL · wang.yixuan@ufl.edu · (352) 888-9238 · nomushroomw.github.io · Google Scholar · GitHub

RESEARCH INTERESTS

Generative modeling; machine learning theory; safety in machine learning; nonlinear control and Lyapunov analysis; reinforcement learning.

EDUCATION

University of Florida — Ph.D., Mechanical & Aerospace Engineering — 2024 – Present
Nonlinear Controls and Robotics (NCR) Lab · Advisor: Prof. Warren E. Dixon · Coadvisor: Prof. Dan P. Guralnik

Johns Hopkins University — M.S., Computer Science — 2023 – 2024

University of Florida — M.S., Mechanical Engineering — 2020 – 2022

Tongji University — B.S., Mechanical Design, Manufacture and Automation — 2015 – 2020

PEER-REVIEWED PUBLICATIONS

1. **Effective Model Pruning: Measuring the Redundancy of Model Components**
   Yixuan Wang, Dan Guralnik, Saiedeh Akbari, Warren E. Dixon
   ICML 2026 **Spotlight** · arXiv:2509.25606
2. **Riemannian Lyapunov Framework: Optimization as Closed-Loop Control on Riemannian Manifolds**
   Yixuan Wang, Omkar Sudhir Patil, Warren E. Dixon
   NeurIPS 2026 · arXiv:2601.22284
3. **Quantifying Trade-Offs Between Stability and Goal-Obfuscation**
   Yixuan Wang, Dan Guralnik, Warren E. Dixon
   IEEE Conference on Decision and Control (CDC) 2026 · arXiv:2605.06630
4. **Goal Inference with Rao-Blackwellized Particle Filters**
   Yixuan Wang, Dan P. Guralnik, Warren E. Dixon
   IFAC World Congress 2026 · arXiv:2512.09269
5. **Off-Dynamics Reinforcement Learning via Domain Adaptation and Reward Augmented Imitation**
   Yihong Guo, **Yixuan Wang**, Yuanyuan Shi, Pan Xu, Anqi Liu
   NeurIPS 2024 · arXiv:2411.09891

PREPRINTS & MANUSCRIPTS

1. **Skeletons in Flow: Graph Structured Flow Matching for Human Motion Prediction**
   Yixuan Wang, Brandon C. Fallin, Warren E. Dixon
   arXiv preprint, 2026 · arXiv:2609.32231
2. **Learn What's Left, Not What's Mastered: Saturation Aware Advantage Reweighting for Multi-Reward Policy Optimization**
   Yixuan Wang, Yifei Chen, Haichao Zhang, Haozheng Luo, Xander Wu, Jie Ni, Yun Fu, Nuno Vasconcelos, Yijiang Li
   arXiv preprint, 2026 · arXiv:2608.16072
3. **Energy Generative Modeling: A Lyapunov-based Energy Matching Perspective**
   Yixuan Wang, Wenqian Xue, Warren E. Dixon
   arXiv preprint, 2026 · arXiv:2605.05530

4. **Lyapunov-Based Kolmogorov-Arnold Network (KAN) Adaptive Control**
   Xuehui Shen, Wenqian Xue, **Yixuan Wang**, Warren E. Dixon
   arXiv preprint, 2025 · arXiv:2512.21437

RESEARCH EXPERIENCE

University of Florida — Graduate Research Assistant, NCR Lab 2024 – Present

Advisor: Prof. Warren E. Dixon · Coadvisor: Prof. Dan P. Guralnik

- **Learning dynamics, geometry, and optimization.** Investigate the dynamics and manifold structure of learning algorithms using nonlinear control and Lyapunov analysis, with applications to optimizer design, reinforcement learning, and stability analysis.
- **Generative modeling and control.** Study generative models and large language models from a control and dynamical systems perspective, including distributional transport, stability, and constrained generation.
- **Model efficiency.** Study principled measures of redundancy in neural network components and their use in pruning and compression.
- **Inference and autonomous systems.** Develop Bayesian intent inference and control policies that jointly regulate physical motion and an observer's belief, with formal bounds on goal information disclosure.

Johns Hopkins University — Graduate Researcher 2023 – 2024

- **Off-dynamics reinforcement learning.** Policy transfer under dynamics shift via domain-adaptation reward modification and imitation learning (NeurIPS 2024).

RESEARCH PROPOSAL DEVELOPMENT

Contributed to advisor-led proposals funded by the National Science Foundation (NSF), Air Force Office of Scientific Research (AFOSR), and Air Force Research Laboratory (AFRL).

- **Research aim development.** Led the development and writing of 1-2 research aims per proposal, from research questions and technical innovations to technical approaches and research plans.
- **Preliminary research and proposal authorship.** Conducted preliminary experiments and authored the corresponding proposal sections to establish the proposed aims' motivation and feasibility.

PROFESSIONAL SERVICE

Conference Reviewer: ICLR, ICML, NeurIPS 2025 - Present

HONORS & AWARDS

ICML Spotlight — *Effective Model Pruning* 2026

Herbert Wertheim College of Engineering Achievement Award Scholarship, University of Florida 2020

TECHNICAL SKILLS

Machine Learning	PyTorch; generative models (energy-based, diffusion, flow matching); reinforcement learning; model compression
ML Systems	Multi-GPU distributed training (DDP), Slurm-based HPC workflows
Theory	Nonlinear & adaptive control, Lyapunov analysis, dynamical systems, optimization on manifolds, Bayesian estimation & particle filtering
Programming	Python, C/C++, MATLAB / Simulink, Git, LaTeX
Languages	English (proficient), Mandarin (native), German (B2)